Discrete Mathematics
Lecture-1
Logic

Question. What is Discrete Mathematics?

Discrete Mathematics is a branch of mathematics involving discrete elements that uses algebra and arithmetic. It is increasingly being applied in the practical fields of mathematics and computer science. It is a very good tool for improving reasoning and problem-solving capabilities.

Question. What are the categories of Mathematics?

Mathematics can be broadly classified into two categories-

Continuous Mathematics- It is based upon continuous number line or the real numbers. It is characterized by the fact that between any two numbers, there are almost always an infinite set of numbers. For example, a function in continuous mathematics can be plotted in a smooth curve without breaks.

Ex. Time, Temperature, Weight, Distance etc.

Discrete Mathematics- It involves distinct values; i.e. between any two points, there are a

countable number of points. For example, if we have a finite set of objects, the function can be defined as a list of ordered pairs having these objects, and can be presented as a complete list of those pairs.

Ex. Number of students, Purchasing tickets, text messages etc.

Mathematical Logic:

The rules of mathematical logic specify methods of reasoning mathematical statements. Greek philosopher, Aristotle, was the pioneer of logical reasoning. Logical reasoning provides the theoretical base for many areas of mathematics and consequently computer science. It has many practical applications in computer science like design of computing machines, artificial intelligence, definition of data structures for programming languages etc.

Propositional Logic is concerned with statements to which the truth values, "true" and "false", can be assigned. The purpose is to analyze these statements either individually or in a composite manner.

Prepositional Logic - Definition

A proposition is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both. A propositional consists of propositional variables and connectives. We denote the propositional variables by capital letters (A, B, etc). The connectives connect the propositional variables.

Some examples of Propositions are given below -

"Man is Mortal", it returns truth value "TRUE"

"12 + 9 = 3 – 2", it returns truth value "FALSE" The

followings are not a Proposition -

- "A is less than 2". It is because unless we give a specific value of A, we cannot say whether the statement is true or false.
- What time is it? (Questions)
- Read this carefully. (Commands / Imperative) □ Do your homework.
- $x + 1 = 2$. (Non constant values / value is not defined)
- Bangladesh and India (Not statements)
- Turin is the best lecturer. (Opinion)
- He is a college student. (Person is not defined)

Propositional variables / Statement variables

□ Today is Friday. p= Today is Friday. □
It is raining. q= It is raining.

The truth value of the propositional variable can be true or false.

Compound proposition

Compound proposition is a proposition formed by combining two or more simple proposition.

Connectives

The logical operators that are used to form compound proposition is called connectives. In propositional logic generally we use five connectives which are –

Symbol	Math Name	English Name
¬	Negation	NOT
∨	Disjunction	OR
∧	Conjunction	AND
⊕	ExOR	OR..but not both
→	Implication	If …. then
⇔	Equivalence/Biconditional	If & only if

Negation (¬): Let A be a proposition. The *negation of A*, denoted by $\neg A$ (also denoted by $\bar{A}$), is the statement "It is not the case that A." The negation of A is the statement "It is not the case that A." The truth table is as follows -

A	$\neg A$
True	False
False	True

Ex. Find the negation of the proposition "Shahin's PC runs Linux." and express this in simple English.

Solution: The negation is

"It is not the case that Shahin's PC runs Linux"

This negation can be more simply expressed as

"Shahin's PC does not run Linux."

Disjunction /OR (∨): Let A and B be propositions. The *disjunction* of A and B, denoted by $A \vee B$, is the proposition "A or B." The disjunction $A \vee B$ is false when both A and B are false and is true otherwise. The truth table is as follows-

A	B	$A \vee B$
True	True	True
True	False	True
False	True	True
False	False	False

Conjunction /AND (∧): Let A and B be propositions. The *conjunction* of A and B, denoted by $A \wedge B$, is the proposition "A and B." The conjunction $A \wedge B$ is true when both A and B are true and is false otherwise. The truth table is as follows –

A	B	$A \wedge B$
True	True	True
True	False	False
False	True	False
False	False	False

Exclusive or/ ExOR: Let A and B be propositions. The exclusive or of A and B, denoted by $A \oplus B$, is the proposition that is true when exactly one of A and B is true and is false otherwise. The truth table is as follows-

A	B	$A \oplus B$
True	True	False
True	False	True
False	True	True
False	False	False

Conditional Statement / Implication / if-then (→): Let A and B be propositions. The conditional statement $A \rightarrow B$ is the proposition "if A, then B." The conditional statement A

→B is false when A is true and B is false, and true otherwise. In the conditional statement A → B, A is called the hypothesis (or antecedent or premise) and B is called the conclusion (or consequence). The truth table is as follows-

A	B	A → B
True	True	True
True	False	False
False	True	True
False	False	True

Biconditional Statement / If and only if (⇔): Let A and B be propositions. The biconditional statement A ↔ B is the proposition "A if and only if B." The biconditional statement A ↔ B is true when A and B have the same truth values, and is false otherwise. Biconditional statements are also called bi-implications. The truth table is as follows-

A	B	$A \to B$
True	True	True
True	False	False
False	True	False
False	False	True

Tautologies:

A Tautology is a formula which is always true for every value of its propositional variables.

Example: Prove $[(A \to B) \wedge A] \to B$ is a tautology.

Proof: The truth table is as follows-

A	B	$A \to B$	$(A \to B) \wedge A$	$[(A \to B) \wedge A] \to B$
True	True	True	True	True
True	False	False	False	True
False	True	True	False	True
False	False	True	False	True

As we can see every value of $[(A \to B) \wedge A] \to B$ is "True", it is a tautology.

Contradictions:

A Contradiction is a formula which is always false for every value of its propositional variables.

Example: Prove $(A \vee B) \wedge [(\neg A) \wedge (\neg B)]$ is a contradiction.

Proof: The truth table is as follows-

A	B	$A \vee B$	$\neg A$	$\neg B$	$(\neg A) \wedge (\neg B)$	$(A \vee B) \wedge [(\neg A) \wedge (\neg B)]$
True	True	True	False	False	False	False
True	False	True	False	True	False	False
False	True	True	True	False	False	False
False	False	False	True	True	True	False

As we can see every value of $(A \vee B) \wedge [(\neg A) \wedge (\neg B)]$ is "False", it is a contradiction.

Contingency:

A Contingency is a formula which has both some true and some false values for every value of its propositional variables.

Example: Prove $(A \vee B) \wedge (\neg A)$ a contingency

Proof: The truth table is as follows-

A	B	$(A \vee B)$	$\neg A$	$(A \vee B) \wedge (\neg A)$
True	True	True	False	False
True	False	True	False	False
False	True	True	True	True
False	False	False	True	False

As we can see every value of $(A \vee B) \wedge (\neg A)$ has both "True" and "False", it is a contingency.

Propositional Equivalences:

Two statements X and Y are logically equivalent if any of the following two conditions hold-

- The truth tables of each statement have the same truth values.
- The bi-conditional statement $X \Leftrightarrow Y$ is a tautology.

Example: Prove $\neg(A \vee B)$ and $[(\neg A) \wedge (\neg B)]$ are equivalent.

Proof: The truth table is as follows-

A	B	$A \vee B$	$\neg(A \vee B)$	$\neg A$	$\neg B$	$(\neg A) \wedge (\neg B)$
True	True	True	False	False	False	False
True	False	True	False	False	True	False

False	True	True	False	True	False	False
False	False	False	True	True	True	True

Here, we can see the truth values of $\neg(A \vee B)$ and $[(\neg A) \wedge (\neg B)]$ are same, hence the statements are equivalent.

Inverse, Converse, and Contra-positive:

Implication / if-then ($\rightarrow$) is also called a conditional statement. It has two parts-

- Hypothesis, p
- Conclusion, q

As mentioned earlier, it is denoted as $p \rightarrow q$

Example of Conditional Statement- "If you do your homework, you will not be punished." Here, "you do your homework" is the hypothesis, p, and "you will not be punished" is the conclusion, q.

Inverse- An inverse of the conditional statement is the negation of both the hypothesis and the conclusion. If the statement is "If p, then q", the inverse will be "If not p, then not q". Thus, the inverse of $p \rightarrow q$ is $\neg p \rightarrow \neg q$.

Example- The inverse of "If you do your homework, you will not be punished" is "If you do not do your homework, you will be punished."

Converse- The converse of the conditional statement is computed by interchanging the hypothesis and the conclusion. If the statement is "If p, then q", the converse will be "If q, then p". The converse of $p \rightarrow q$ is $q \rightarrow p$.

Example- The converse of "If you do your homework, you will not be punished" is "If you will not be punished, you do your homework".

Contra-positive- The contra-positive of the conditional is computed by interchanging the hypothesis and the conclusion of the inverse statement. If the statement is "If p, then q", the contra-positive will be "If not q, then not p". The contra-positive of $p \rightarrow q$ is $\neg q \rightarrow \neg p$.

Example- The Contra-positive of "If you do your homework, you will not be punished" is "If you are punished, you did not do your homework".

Exercise

Construct the truth table for each of these compound propositions

(1) $(A \vee \neg B) \rightarrow (A \wedge B)$.
(2) $(A \oplus A)$
(3) $(A \oplus \neg A)$
(4) $(A \oplus \neg B)$
(5) $(\neg A \Leftrightarrow \neg B) \Leftrightarrow (A \Leftrightarrow B)$